

A SUMMARY OF THE ENDOMORPHISM CONJECTURE ON FANO MANIFOLDS OF PICARD NUMBER 1

GUOLEI ZHONG

We work over the field $\mathbb{C}$ of complex numbers.

0.1. Endomorphism Conjecture.

Conjecture 0.1. Let X be a Fano manifold of Picard number 1. If X admits a non-isomorphic surjective endomorphism, then X is a projective space.

Remark 0.2. Conjecture 0.1 has been verified in the following cases:

- (1) Almost homogeneous spaces ([PS89], [HN11]);
- (2) Smooth hypersurfaces of a projective space ([PS89], [Bea01]; cf. [KT23]);
- (3) Fano threefolds ([ARVdV99], [HM03]; cf. [KT23]);
- (4) Fano manifolds containing a rational curve with trivial normal bundle ([HM03]);
- (5) Fano fourfolds with Fano index ≥ 2 ([SZ24]; cf. [KT23]); and
- (6) Del Pezzo manifolds, i.e., the Fano index $= \dim(X) - 1$ ([SZ24], [AKP08]).
- (7) Mukai manifolds, i.e., the Fano index $= \dim(X) - 2$ ([SZ23, SZ24]; cf. [KT23]).
- (8) The tangent bundle T_X is big and the VMRT along a general point is not dual defective ([SZ23]).

0.2. BDP Conjecture.

Definition 0.3 (BDP). Let X be a Fano manifold of Picard number 1. We say that X satisfies *bounded-degree-property* (BDP for short) if for any smooth projective variety Y of Picard number 1, the degree of $f: Y \rightarrow X$ (if exists) is bounded in terms of the discrete invariants of Y and X .

Definition 0.4 (BVP). Let X be a smooth projective variety. We say that X satisfies the *Bott-vanishing* (BVP for short) if $H^j(X, \Omega_X^i \otimes L) = 0$ for every $j > 0$, $i \geq 0$, and L ample.

Conjecture 0.5 (Van-de-Ven; see [Ame97, Conjecture B]). Any Fano manifold of Picard number 1 different from the projective space satisfies BDP.

Remark 0.6. Clearly, a positive answer to Conjecture 0.5 implies a positive answer to Conjecture 0.1. Conjecture 0.5 has been verified in the following cases:

- (1) Fano threefolds ([Ame97] and [ARVdV99]; cf. [KT23]).

- (2) Smooth hypersurfaces ([SZ24], [Ame07]; cf. [KT23]).
- (3) Fano fourfolds with Fano index ≥ 2 ([KT23]; cf. [SZ24]).
- (4) Del Pezzo manifolds ([SZ24], [KT23]).

It is not hard to check that all of the above manifolds violate BVP. It has been proved in [KT23] that, BDP holds for those X violating BVP.

Question 0.7 (Hwang). Does BVP/BDP hold for Fano manifolds of Picard 1 containing a rational curve with trivial normal bundle?

Remark 0.8. It has been proved in [HM03] that for $f: Y \rightarrow X$ a finite surjective morphism of Fano manifolds of Picard number 1, if X contains a rational curve with trivial normal bundle, then the degree of f is bounded in terms of the discrete invariants of Y and X . So the above question is to extend [HM03] by removing the condition Y being Fano.

0.3. TPP Problem.

Definition 0.9 (TPP). Let X be a Fano manifold of Picard number 1. We say that X satisfies *target- $\mathbb{P}^n$ -property* (*TPP* for short) if for any non-isomorphic surjective morphism $g: X \rightarrow Y$ onto a smooth projective variety Y , we have $Y \cong \mathbb{P}^n$.

Question 0.10 (Shao-Zhong; see [SZ23, Question 1.3]). Any Fano manifold of Picard number 1 whose tangent bundle is big satisfies TPP.

Remark 0.11. Question 0.10 holds in the following situations.

- (1) Rational homogeneous spaces (Lazarsfeld problem=Hwang-Mok theorem [HM99]).
- (2) Fano threefolds ([HL23, Laz84, SZ23]; cf. [Ame04]).

In [SZ23], we proved that, if (1) $g: X \rightarrow Y$ is a non-isomorphic surjective morphism of Fano manifolds of Picard number 1 whose VMRTs are not dual defective, and (2) T_X is big, then Y is a projective space. Therefore, to verify TPP, one needs to remove the condition on the defect of VMRT's.

Remark 0.12. TPP in general does not hold if we remove the condition that T_X is big. Some examples below.

- (1) A del Pezzo manifold X of degree 4 (which is a smooth complete intersection of two quadric hypersurfaces) admits a double cover over a smooth quadric hypersurface (see [SZ24, Proposition 3.6]). The tangent bundle of X is $\mathbb{Q}$ -effective but not big (see [BEH⁺23]).
- (2) Given a multi-degree $\underline{d} = (d_1, \dots, d_n)$ with $d_1 \geq \dots \geq d_n \geq 2$, there is a smooth complete intersection X which admits a degree d_n cover onto a smooth complete

intersection of multi-degree $(d_1, \dots, d_{n-1})$ (see [SZ23, Proof of Theorem 1.5]). The tangent bundle of X is not big, and even not pseudo-effective if $d_1 \geq 3$ (see [SZ23, Theorem 1.5]).

- (3) There is a double cover from a special Gushel-Mukai manifold X to a quintic del Pezzo manifold of degree 5. The tangent bundle of X is not big when $\dim(X) \neq 6$ (see [SZ23, Proposition 1.7]).

Based on the above examples, we ask the following question related to Question 0.10.

Question 0.13. Let X be a Gushel-Mukai sixfold. Is T_X non-big?

Remark 0.14. In this case, the VMRT along a general point is dual defective. So we couldn't apply the criterion we developed in [SZ23, Theorem 1.2]. We believe that the tangent bundle T_X is still non-big. If T_X is big, it would provide a negative answer to Question 0.10 as it admits a double cover onto a quintic del Pezzo sixfold of degree 5.

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GUOLEI ZHONG, CENTER FOR COMPLEX GEOMETRY, INSTITUTE FOR BASIC SCIENCE (IBS), 55 EXPO-RO, YUSEONG-GU, DAEJEON, 34126, REPUBLIC OF KOREA.

Email address: zhongguolei@u.nus.edu, guolei@ibs.re.kr

URL: <https://www.zhong-guolei.com>